

5(a)(i)	U4 – T-helper (lymphocyte / cell) AND V4 – T-killer / T-cytotoxic (lymphocyte / cell) ; A ‘helper and killer’ I helping and killing	1
5(a)(ii)	any four from: <i>U4 / helper</i> 1 secretes / produces / releases, cytokines / interleukins ; I cell signalling molecules / cytokinins 2 to stimulate, B-lymphocytes / B cells, to, divide / develop into plasma cells ; A stimulates humoral response / clonal expansion / development of memory cells 3 stimulate macrophages to, be active / carry out phagocytosis / respond (to presence of pathogens) / be ‘angry’ ; I endocytosis <i>V4 / killer</i> 4 produces / secretes / releases, chemicals, to kill / break down, infected cells ; A (auto)lysis / apoptosis 5 e.g. perforin / hydrogen peroxide / granzymes / proteases ; A hydrolytic enzymes / toxins 6 detail of action of named chemical (see below) ; 7 AVP ; e.g. correct ref to CD4 receptors (on U4 / T-helper) / CD8 receptors on (V4 / T-killer) perforin – makes pores in cell surface membrane of infected cell for delivery of granzymes hydrogen peroxide – oxidises cell components / stimulates apoptosis granzymes / proteases – break down proteins (granzymes are serine proteases)	4

Question	Answer	Marks
5(b)	any four from: 1 make vaccine free ; I cheap 2 make vaccine <u>available</u>, globally / to whole population (urban and rural) ; I vaccinating people 3 <i>ref to</i> herd immunity / described ; e.g. vaccinate a high proportion of the population 4 vaccinate children as early as possible ; 5 <i>ref to</i> provide boosters / use effective vaccine so boosters are not necessary ; e.g. live / attenuated, virus 6 accurate record keeping of those vaccinated / find those who have not been vaccinated ; 7 surveillance / <i>ref to</i> contact tracing / described ; e.g. look out for any cases of the disease 8 start vaccination immediately cases are found / <i>ref to</i> ring immunity / benefits of ring immunity ; 9 education about, effects of the disease / benefits of having the vaccine / transmission of the disease ; A counter, misinformation / antivaxers A importance of vaccination 10 make sure there is a sufficient supply of the vaccine ; 11 provide funding for vaccine, production / distribution / delivery ; A provide, cheap / easy to produce, vaccine 12 provide trained personnel ; 13 AVP ; e.g. quarantining people with the disease evaluate or monitor success of vaccination programme maintain cold chain (to ensure vaccine is effective) use a thermostable vaccine provide different types of the vaccine (e.g. Sabin (oral) and Salk (injected); different manufacturers for Covid) I develop new vaccines in response to new, strains / variants / AW, of pathogen	4